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TÉLÉCOM SUDPARIS
Master of Sciences 2 , Data Science and Network Intelligence

Design and Evaluation of a Retrieval-Augmented Generation (RAG) System Using Large Language Models

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Academic Year: 2025 – 2026

September 19, 2026

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1 Introduction

1.1 Problem Statement

Large language models excel at text generation but frequently produce hallucinations—fluent text that is factually incorrect or unsupported by source material. Retrieval-Augmented Generation (RAG) addresses this by grounding generation in retrieved documents, but vanilla RAG systems still fail to ensure answers remain faithful to retrieved content.

1.2 Objectives

This project addresses two key questions:

1. Can prompt engineering alone achieve high faithfulness without fine-tuning?
2. How does model size and training objective affect self-verification capability?

Task Definition. Given a user query and a set of retrieved documents, the system must:

1. Retrieve relevant context documents
2. Generate an answer using *only* the retrieved content
3. Decompose the generated answer into individual factual claims
4. Verify each claim against the retrieved context
5. Produce a faithfulness score and an overall verdict (PASS / FAIL / PARTIAL)

This task combines elements of generative question answering, hallucination detection, and truthfulness evaluation at the claim level.

2 VeriRAG Architecture

2.1 System Overview

VeriRAG uses a single language model for both generation and verification, unlike multi-agent approaches requiring separate models.

Question → FAISS Retrieval → LLM Self-Verification → Answer

The system operates in three stages:

1. **Document Retrieval:** FAISS retrieves $k = 4$ most relevant chunks using sentence-transformers embeddings
2. **Self-Verification:** A structured prompt instructs the LLM to verify answerability before generation
3. **Evaluation:** Multi-metric scorer measures claim-level faithfulness

2.2 Multi actors Prompt Architecture

We use a single underlying LLM , while structuring the pipeline into multiple roles via prompting :

- **Query Planner:** Determines answerability and retrieval strategy
- **Answer Generator:** Produces answers strictly grounded in retrieved context
- **Claim Verifier:** Extracts and verifies atomic factual claims

Question → Planner → Retriever → Generator → Verifier

This modular design improves interpretability and allows fine-grained error analysis.

2.3 Key Design Choices

2.3.0.1 Zero Training. Prompt engineering avoids fine-tuning costs and maintains reproducibility across different deployments.

2.3.0.2 Explicit Verification. The prompt instructs the model to check answerability and abstain when uncertain, enabling reliable out-of-domain detection.

3 Dataset and Methodology

3.1 Document Corpus

We constructed a corpus from 11 documents on machine learning topics (transformers, neural networks, overfitting, LLMs, etc.)

3.1.0.1 Machine Learning and AI (6 articles):

- Transformer (machine learning model)
- Neural Network
- Machine Learning
- Overfitting
- Large Language Model
- Turing Test

3.1.0.2 Physics (5 articles):

- Black Hole
- General Relativity
- Albert Einstein
- Curved Spacetime
- The Big Bang

Wikipedia was selected due to its high-quality, well-structured content. The inclusion of two distinct domains enables evaluation of cross-domain generalization.

3.1.1 Data Preparation

3.1.1.1 Chunking. Documents were segmented into chunks of 500 words with a 50-word overlap to preserve semantic continuity across boundaries.

3.1.1.2 Embeddings. Each chunk was embedded using the `sentence-transformers/all-MiniLM-L6-v2` model (384 dimensions). A FAISS Flat L2 index was used to ensure exact nearest-neighbor retrieval.

Table 1: Corpus Statistics

Statistic	Value
Total documents	11
Total chunks	237
Average chunk length	487 words
Vocabulary size	~8,500 tokens
FAISS index size	91 KB

3.2 Evaluation Questions

We created 24 questions across three categories:

Table 2: Question Distribution

Category	Count	Purpose
Easy (ML concepts)	11	Test basic retrieval
Medium (Technical)	8	Test understanding
Out-of-domain	5	Test abstention

Out-of-domain questions (baking, sports, geography) test whether the model can refuse to answer when information is unavailable.

3.3 Models Evaluated

Table 3: Language Models Tested

Model	Parameters	Type
Qwen-2.5-1.5B-Instruct	1.5B	Instruction-tuned
Llama-3.2-1B-Instruct	1.0B	Instruction-tuned
Llama-3.2-3B-Instruct	3.0B	Instruction-tuned
Phi-2	2.7B	Code-focused
TinyLlama-1.1B-Chat	1.1B	General chat

4 Faithfulness Scoring

The primary evaluation metric is **faithfulness**, defined as:

$$\text{Faithfulness} = \frac{\text{Number of supported claims}}{\text{Total number of claims}}$$

4.1 Multi-Metric Approach

Our faithfulness scorer combines five complementary metrics to evaluate whether generated answers are supported by retrieved context.

4.1.1 Metrics and Weights

Table 4: Faithfulness Scoring Metrics

Metric	Weight	Purpose
Semantic Similarity	35%	Captures paraphrasing via sentence embeddings
ROUGE-L	25%	Longest common subsequence (preserves order)
ROUGE-2	20%	Bigram overlap (phrasal similarity)
BLEU	10%	N-gram precision (1–4 grams)
Word Overlap	10%	Stemmed token matching (fallback)

4.1.2 Scoring Process

1. **Claim Extraction:** Split answer into sentences; filter claims with ≥ 4 words and $\geq 30\%$ unique words
2. **Claim Verification:** For each claim, compute combined metric score against retrieved context
3. **Support Threshold:** Claim is supported if score ≥ 0.18 (lenient to accept paraphrasing)
4. **Faithfulness:** Ratio of supported claims to total claims

4.1.3 Verdict Assignment

- **PASS:** Faithfulness ≥ 0.40 (40% of claims supported)
- **PARTIAL:** $0.25 \leq \text{Faithfulness} < 0.40$
- **FAIL:** Faithfulness < 0.25
- **ABSTAIN:** Answer contains “I cannot answer” (treated as success)

The 0.18 claim threshold and 0.40 pass threshold were empirically tuned on Llama-3B development runs:

5 Experimental Results

5.1 Overall Performance

Table 5: Main Results Across All Models

Model	Vanilla	VeriRAG	Improvement	Pass Rate
Qwen-1.5B	75.9%	92.1%	+21.4%	100%
Llama-3B	81.3%	92.8%	+14.1%	100%
Llama-1B	81.5%	82.6%	+1.4%	87.5%
Phi-2	81.9%	69.7%	-15.0%	79.2%
TinyLlama	29.2%	12.0%	-17.2%	12.5%

5.2 Performance by Question Difficulty

5.2.1 Llama-3.2-3B (Best Absolute)

Table 6: Llama-3B Performance Breakdown

Category	Vanilla	VeriRAG	Change
Easy (11 questions)	74.6%	93.5%	+18.9%
Medium (8 questions)	94.8%	91.2%	-3.6%

VeriRAG significantly improves easy questions but shows slight regression on medium questions where vanilla already performed well.

5.2.2 Qwen-2.5-1.5B (Best Improvement)

Table 7: Qwen-1.5B Performance Breakdown

Category	Vanilla	VeriRAG	Change
Easy (11 questions)	66.7%	90.5%	+23.8%
Medium (8 questions)	94.3%	95.3%	+1.0%

Qwen’s lower baseline (75.9%) provides more room for improvement, achieving the largest absolute gain (+21.4%).

6 Analysis and Discussion

6.1 Key Findings

6.1.1 Finding 1: Instruction-Tuning Matters More Than Size

Table 8: Size vs Training Objective Impact

Model	Size	Training Focus	VeriRAG Δ
Qwen-1.5B	1.5B	Instruction-following	+21.4%
Phi-2	2.7B	Code generation	-15.0%

Despite being 80% larger, Phi-2 performs 36.4 percentage points worse than Qwen-1.5B with VeriRAG. This demonstrates that instruction-tuning capability is more critical than parameter count.

6.1.1.1 Root Cause (Phi-2 Failure). Analysis of Phi-2 outputs reveals:

- Poor abstention: Only 1/5 out-of-domain questions refused (vs 5/5 for Qwen/Llama-3B)
- Low confidence scores: Average 0.247 (below 0.40 PASS threshold)
- Weak instruction-following: Phi-2 trained primarily on code, not meta-instructions

6.1.2 Finding 2: Baseline-Improvement Inverse Correlation

Table 9: Baseline Performance vs VeriRAG Gain

Model	Vanilla Baseline	VeriRAG Gain
Qwen-1.5B	75.9% (weak)	+21.4% (large)
Llama-3B	81.3% (medium)	+14.1% (medium)
Llama-1B	81.5% (strong)	+1.4% (small)

VeriRAG provides greatest value for models with weaker baselines. Well-calibrated models (Llama-1B at 81.5%) show minimal improvement because they already generate grounded responses.

6.2 Comparison to Literature

Table 10: Comparison with Related Work

Method	Faithfulness	Training	Inference	Size
Vanilla RAG	75–81%	None	1×	Any
Self-RAG (Asai et al.)	90–95%	Required	1.5×	7B+
Multi-Agent RAG	88–93%	None	3×	3B+ (×3)
VeriRAG (Llama-3B)	92.8%	None	3×	3.0B
VeriRAG (Qwen-1.5B)	92.1%	None	3×	1.5B

6.3 Out-of-Domain Abstention

Table 11: Abstention Accuracy on Out-of-Domain Questions

Model	Vanilla	VeriRAG
Qwen-1.5B	0% (0/5)	100% (5/5)
Llama-3B	0% (0/5)	100% (5/5)
Llama-1B	0% (0/5)	40% (2/5)
Phi-2	0% (0/5)	20% (1/5)

Perfect abstention for instruction-tuned models demonstrates that VeriRAG prevents hallucination on unanswerable queries.

7 Limitations and Future Work

7.1 Current Limitations

Despite its effectiveness, the proposed approach presents several limitations. First, VeriRAG relies on instruction-tuned language models capable of following structured prompts and performing self-verification; base pretrained models or code-focused models, such as Phi-2, exhibit degraded performance under this framework. Second, the evaluation was restricted to English-language documents and primarily focused on the machine learning and artificial intelligence domain, limiting conclusions about cross-lingual or cross-domain generalization. Third, experiments were conducted only on models with up to 3B parameters, leaving the behavior of larger-scale models (7B and above) unexplored. Finally, the verification mechanism depends on empirically tuned faithfulness thresholds, and these values may require recalibration when applied to different domains or datasets, particularly under distributional shifts.

7.2 Future Improvements

Several extensions of this work can be explored. First, evaluating VeriRAG on larger-scale language models (7B parameters and beyond, such as Llama-3-8B or Mixtral) would help determine whether explicit verification remains beneficial at higher reasoning capacities or

whether such models are already sufficiently well-calibrated. Second, domain adaptation represents a promising direction, particularly for high-stakes domains such as medicine or law, where prompt structures could be tailored to domain-specific reasoning patterns and evidence requirements. Third, extending VeriRAG to multi-turn conversational settings would enable the study of faithfulness preservation across dialogue history, a critical challenge for interactive systems. Finally, future work could investigate the synergy between verification-aware prompting and parameter-efficient fine-tuning, comparing the effectiveness of VeriRAG alone with fine-tuned models augmented by verification prompts.

8 Conclusion

This project introduced **VeriRAG**, a verification-aware retrieval-augmented generation framework implemented using a **single underlying LLM** and **multi-stage prompting** (planning, grounded answering, and claim-level verification). Without any fine-tuning, VeriRAG significantly improved faithfulness for instruction-tuned models, notably **Llama-3.2-3B** ($0.813 \rightarrow 0.928$) and **Qwen-2.5-1.5B** ($0.759 \rightarrow 0.921$), while also enabling reliable abstention on out-of-domain questions (5/5 abstentions for both models).

However, results also highlight that verification is **model-dependent**: non-instruction-tuned or code-focused models such as **Phi-2** degraded under the same prompting strategy ($0.819 \rightarrow 0.697$), suggesting that self-verification requires strong instruction-following and meta-reasoning capabilities. Overall, The project demonstrates that prompt engineering alone can provide a practical and reproducible path to more trustworthy RAG systems, while exposing clear trade-offs in latency and threshold sensitivity that should be addressed in future work.

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